

Visual Design

Gestalt Principles

Creating Organization and Structure

Typography

UI Visual Design Objectives

1. **Information communication**

- Enforce desired relationships (and avoid undesired relationships)

2. **Aesthetics**

- well designed, complete, well ordered, professional, attractive

3. **“Brand”**

- recognizable as being part of your organization

UI Visual Design Problem

- You need to present elements of your interface to your users
- Those users need to know:
 - What can I do in this interface?
 - Where is _____ ?
 - How can I do _____ ?
 - What is related to what?
- Want to impose **as little thinking as possible** on your users
 - Let them concentrate on their task, not the interface

Organization and Structure

- Structure doesn't usually occur naturally, it must be designed
- People will find structure, even if none was intended
 - (and it may be the wrong structure)
- Use Gestalt principles to create organization and structure



Gestalt Principles

- Theories of visual perception that describe how people tend to organize visual elements into groups or unified wholes, when certain principles are applied.
- Clues about how the brain groups raw visual input

Proximity

Similarity

Continuity

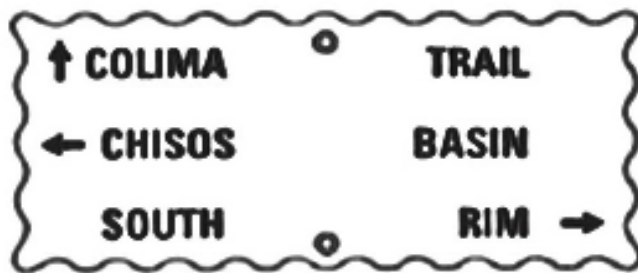
Closure

Connectedness

Figure and Ground

Proximity

- Individual elements are associated more strongly with nearby elements than with those further away
- Best way of emphasizing relationships between different data entities is to place them in proximity in a display



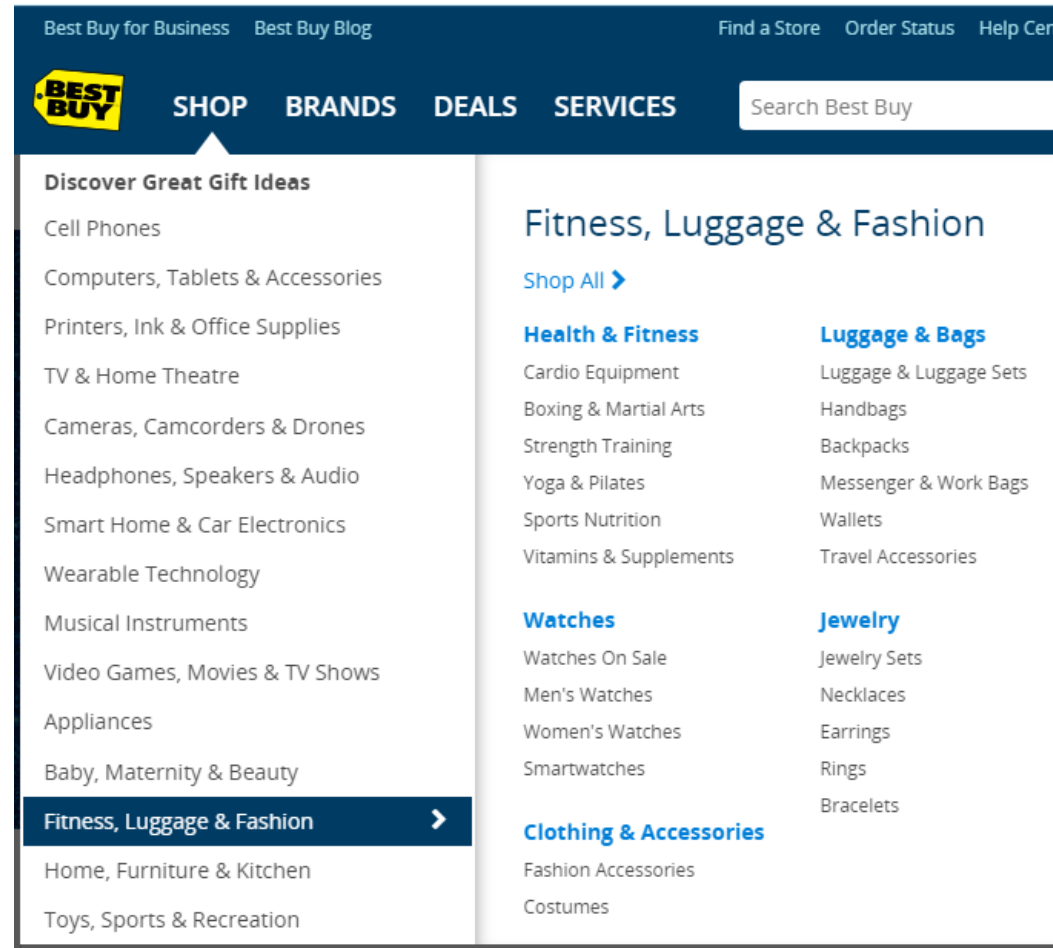
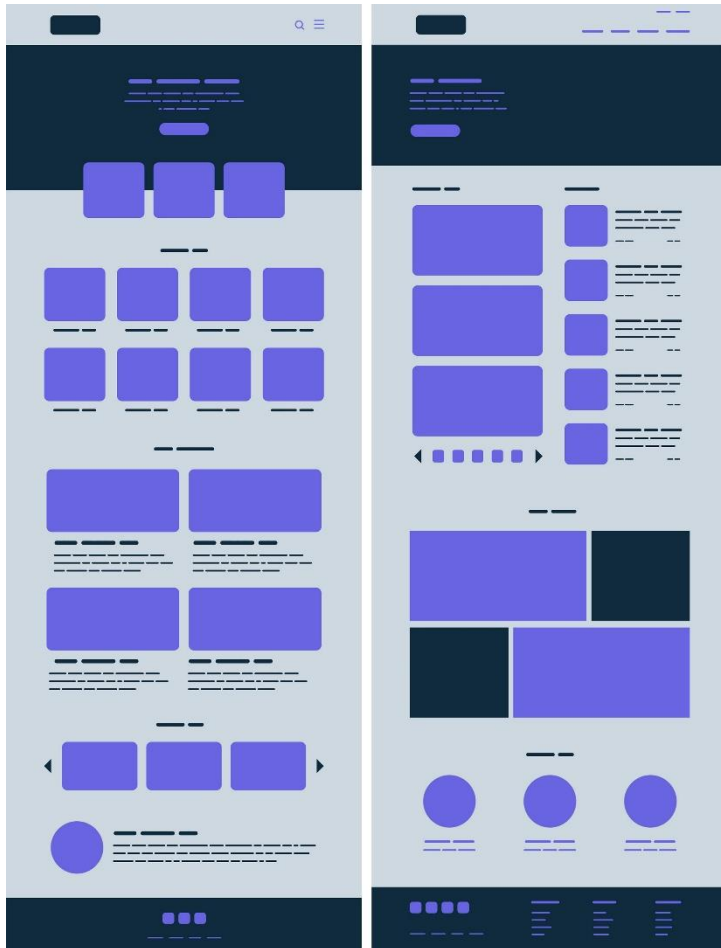
Horizontal



Vertical

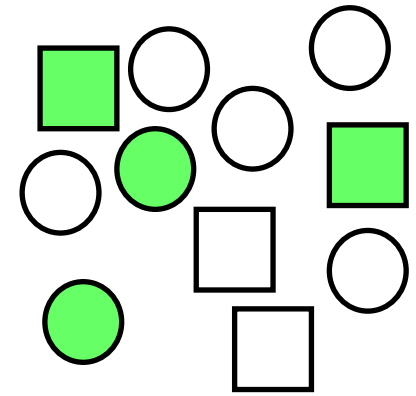
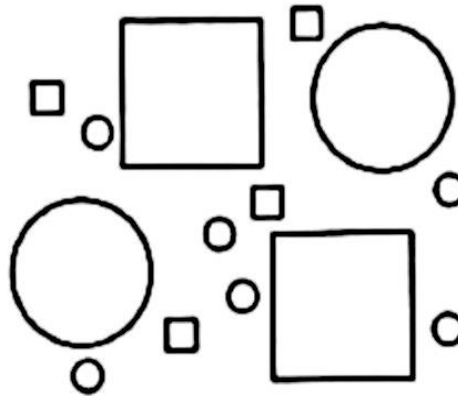
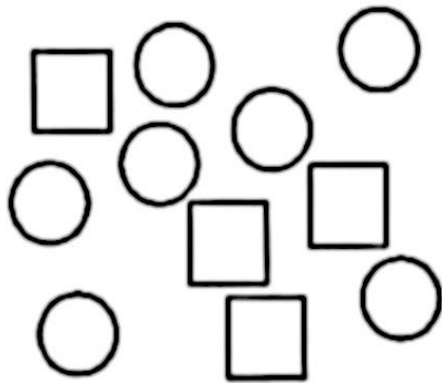
Proximity

- Things that are related should be grouped together
- Things that are not related should be separated from each other

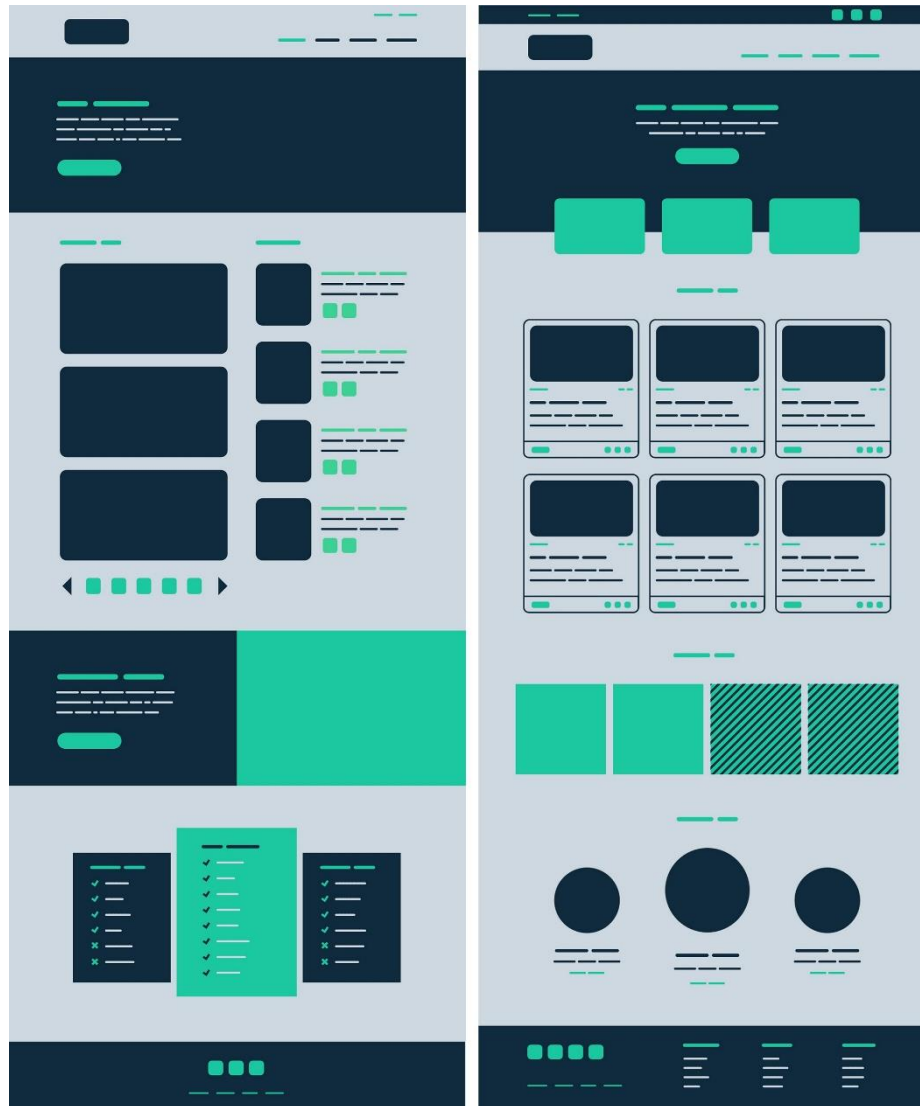


Similarity

- Elements associated more strongly when they share basic visual characteristics, such as:
 - Shape, Size, Color, Texture, Orientation
- Related visual elements should look similar

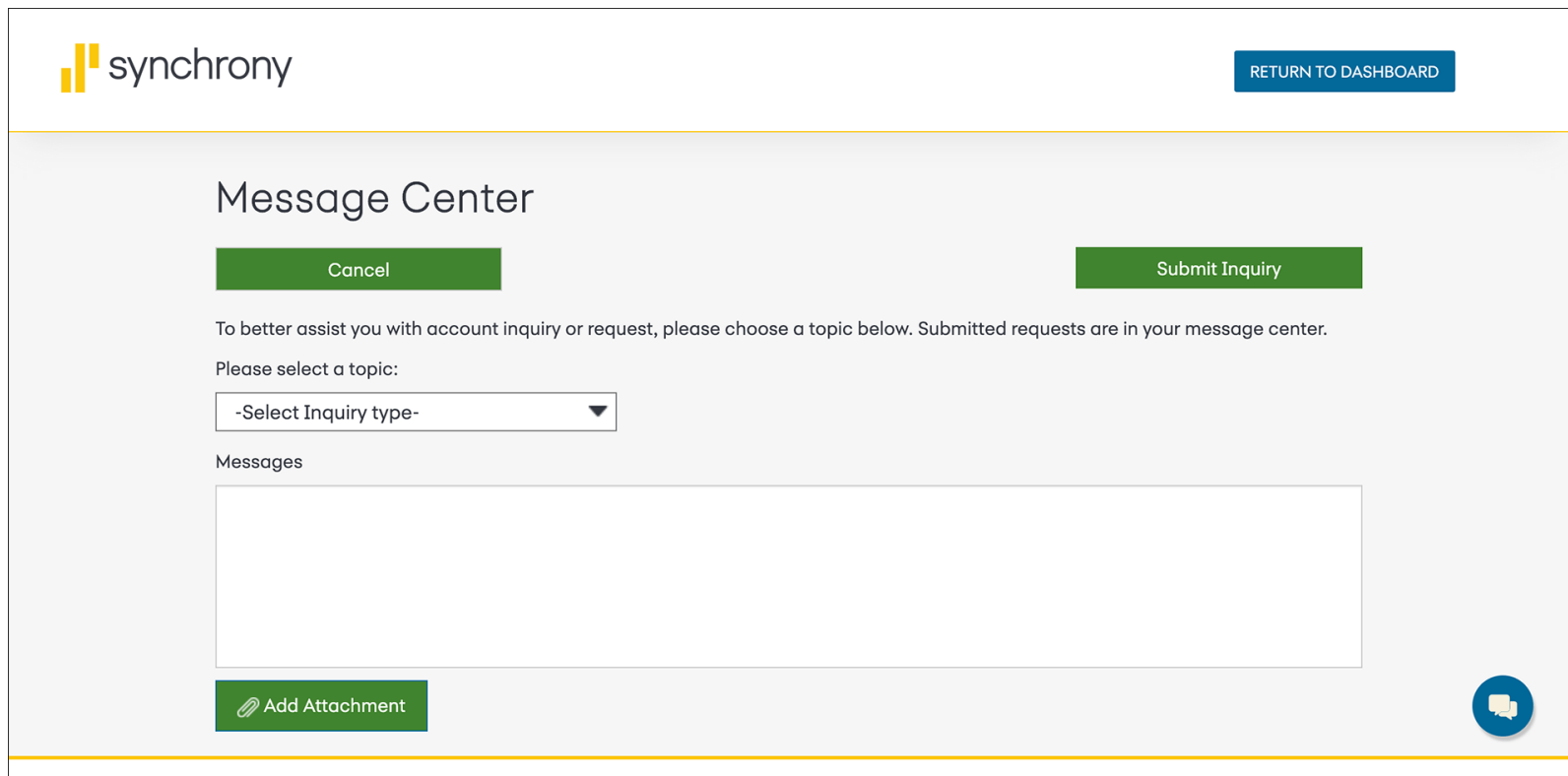


Similarity



Similarity

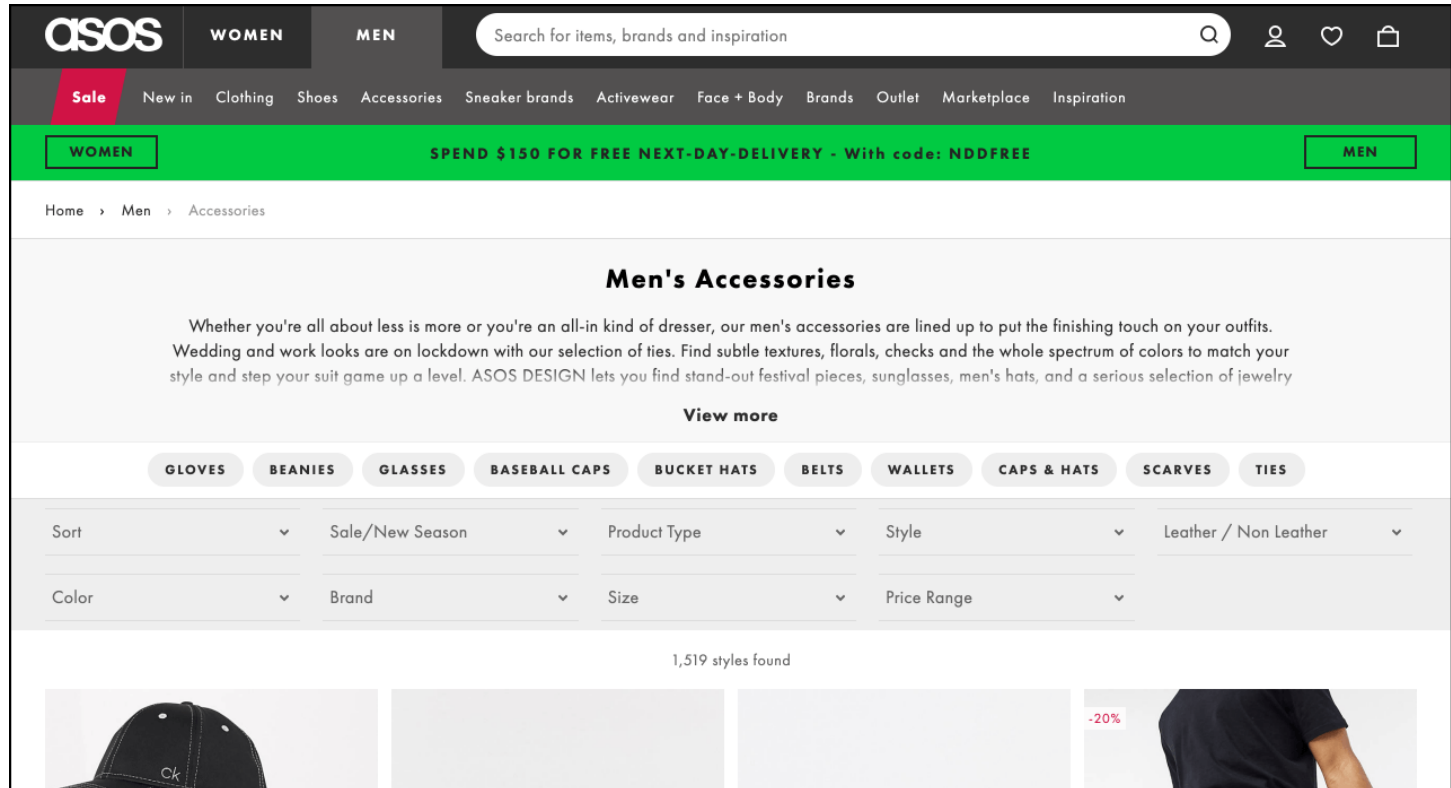
- All the buttons in the Message Center were green, so the user had to spend time figuring out which was the primary button that submitted the form



The screenshot shows the Synchrony Message Center interface. At the top left is the Synchrony logo, and at the top right is a blue button labeled "RETURN TO DASHBOARD". The main heading is "Message Center". Below this are two green buttons: "Cancel" on the left and "Submit Inquiry" on the right. A message states: "To better assist you with account inquiry or request, please choose a topic below. Submitted requests are in your message center." Below this is a prompt "Please select a topic:" followed by a dropdown menu with the text "-Select Inquiry type-". Underneath is a section labeled "Messages" with a large, empty white rectangular box for the message content. At the bottom left is a green button with a paperclip icon labeled "Add Attachment". At the bottom right is a blue circular chat icon.

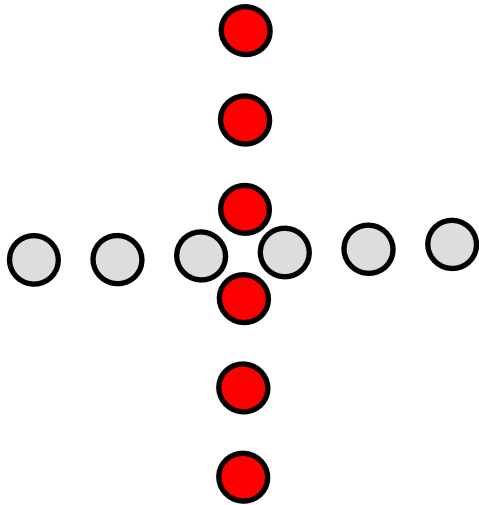
Similarity

- All rounded rectangles strengthen their perception as a group and communicate their similar functionality as navigation.
- The rectangular shape of the filters below signaled that group as having a different functionality.

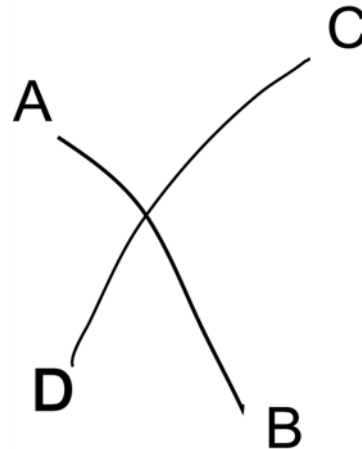


Continuity

- Elements arranged in a straight line or a smooth curve are perceived as being more related
 - bias to continuous forms rather than disconnected segments



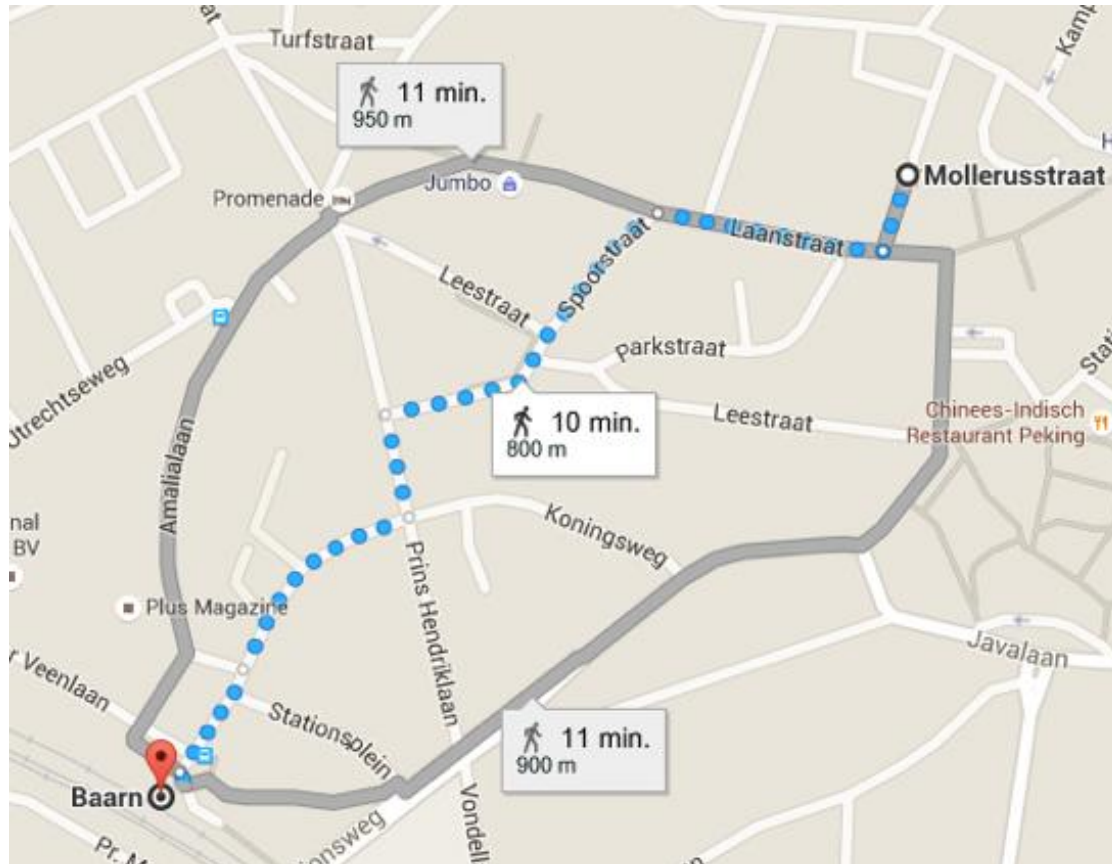
We see two rows of circles, not two L-shaped groups



We see curves AB and CD, not AC and DB, and not AD and BC

Continuity

- Google maps shows walking paths as a series of blue dots which we interpret as a path in the direction of a destination

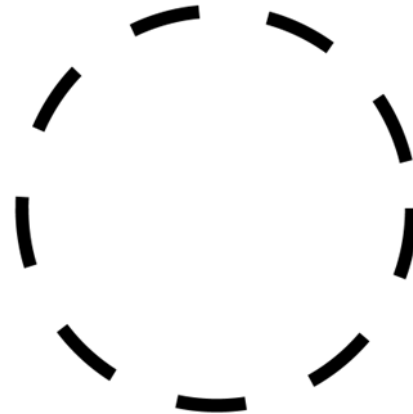


Closure

- The visual system perceives a set of individual elements as a single, recognizable pattern, rather than individual elements.



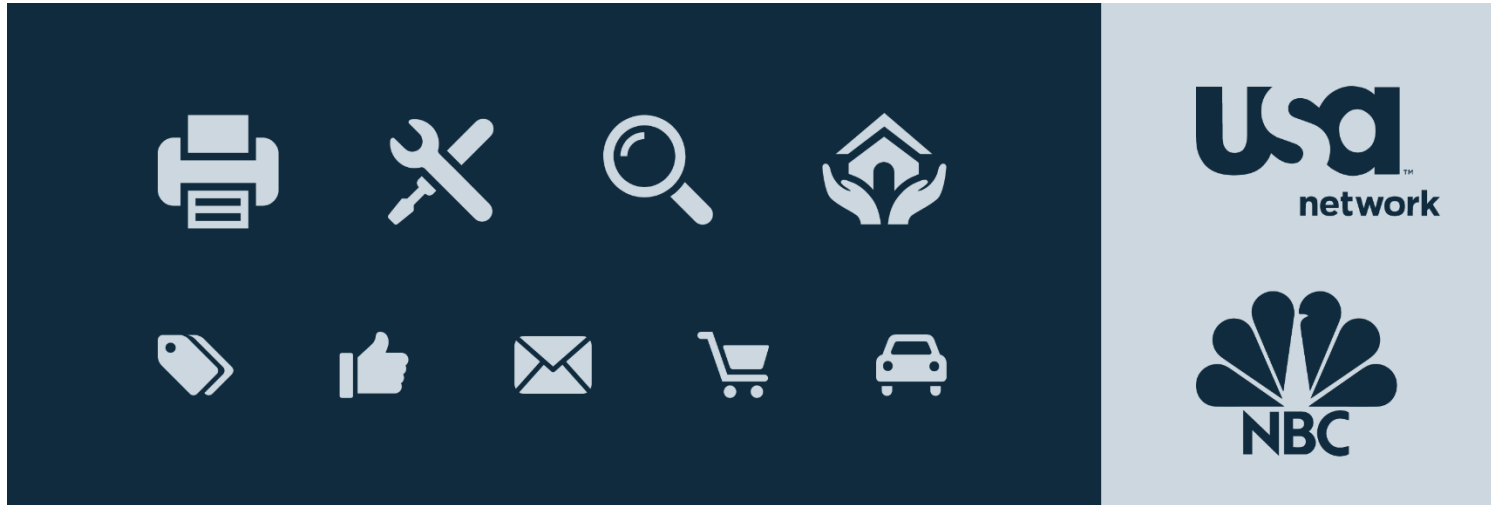
Blobs → Pandas



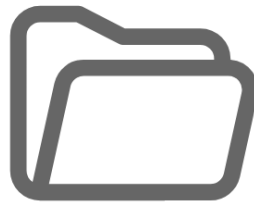
Curved dashed lines → circle

Closure

- The closure principle is commonly used in logo and icon design at a variety of companies



Database



Folder

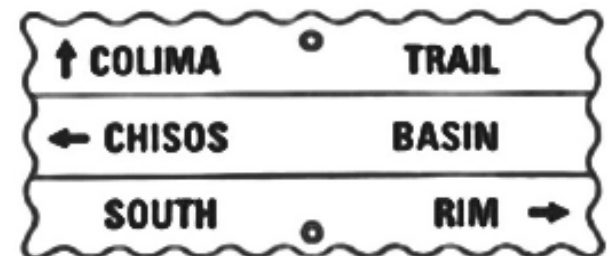
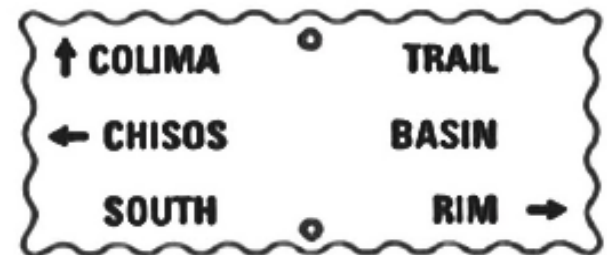
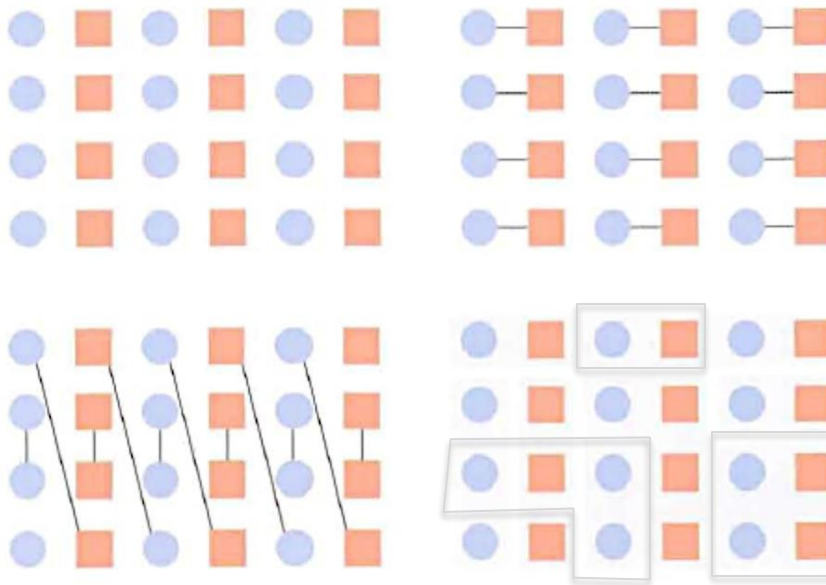


Stack



Connectedness

- Elements connected to one another by uniform visual properties are perceived to be more related than elements that are not connected
- Two typical strategies:
 - connecting lines
 - connecting regions



Connectedness

- The principle of connection is applied in many (public) transport maps, e.g. in metro maps
- The connection between the dots, together with the use of colors, shows which stations are connected.
- Without the lines, this map would be worthless

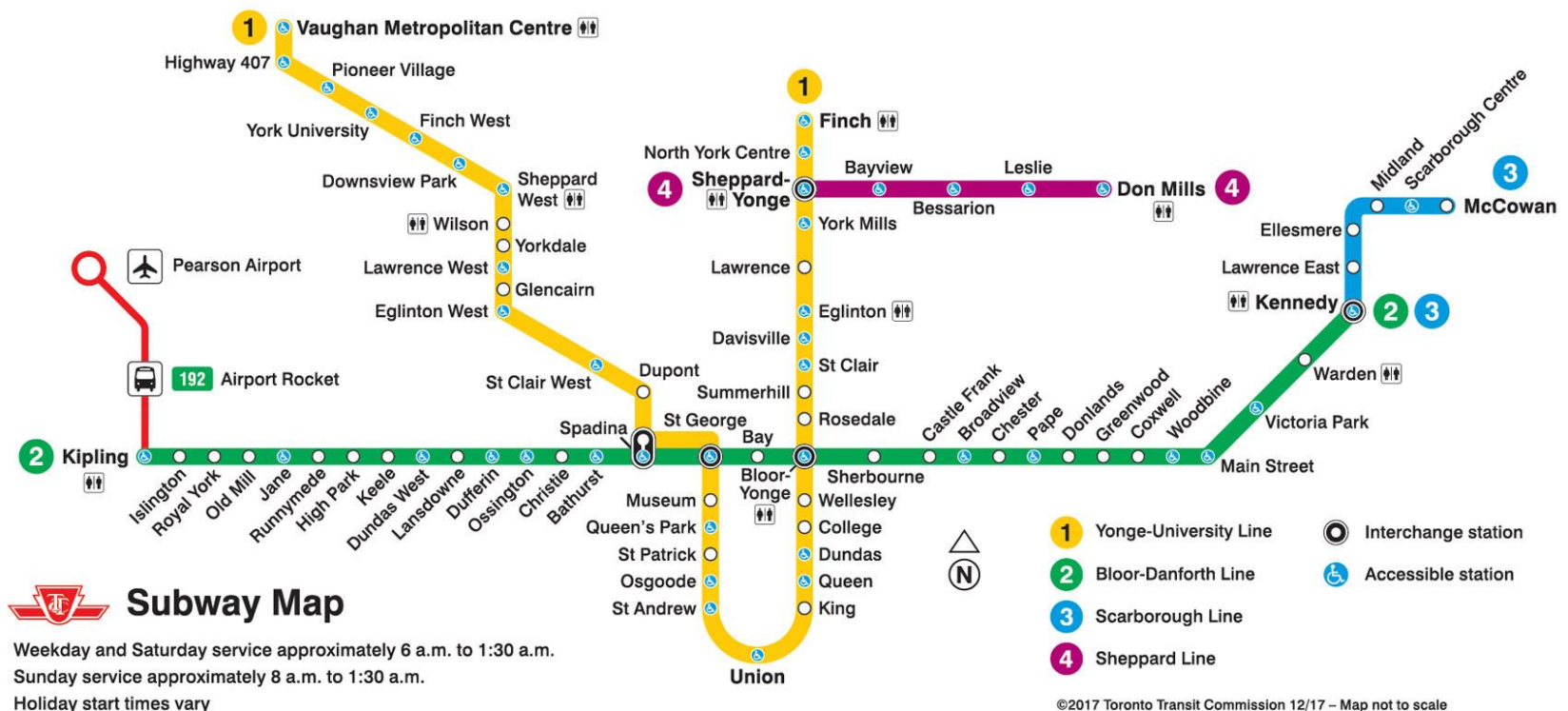


Figure and Ground

- Elements are perceived as either figure i.e., the element of focus or ground i.e., background on which the figure sits.
- Brain will interpret the larger area of an image as the ground and the smaller as the figure
- This principle essentially tells our eyes what to focus on first

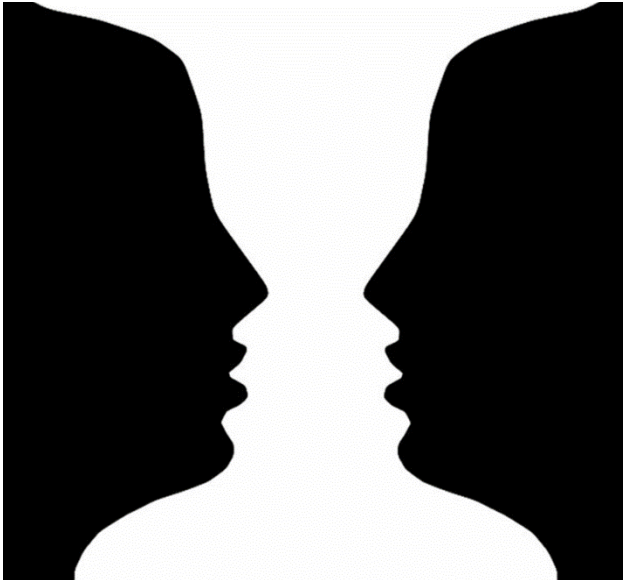


Figure and Ground

- Things get interesting when the foreground and background actually contain two distinct images
- Techniques to distinguish: you can use parallax background, semi-transparent overlay, shadows or blur the elements in the background.

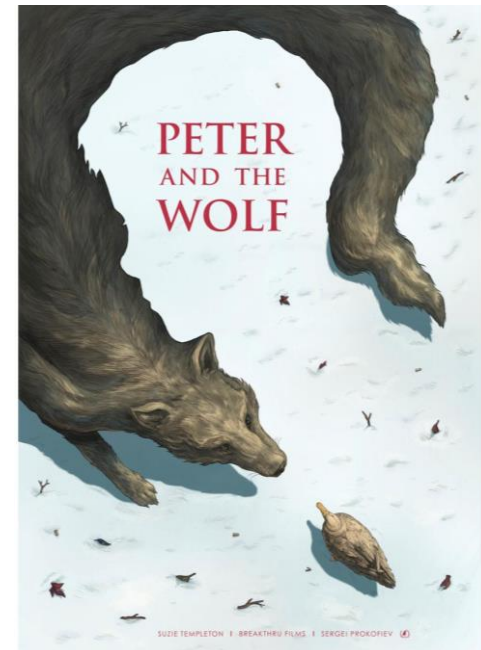
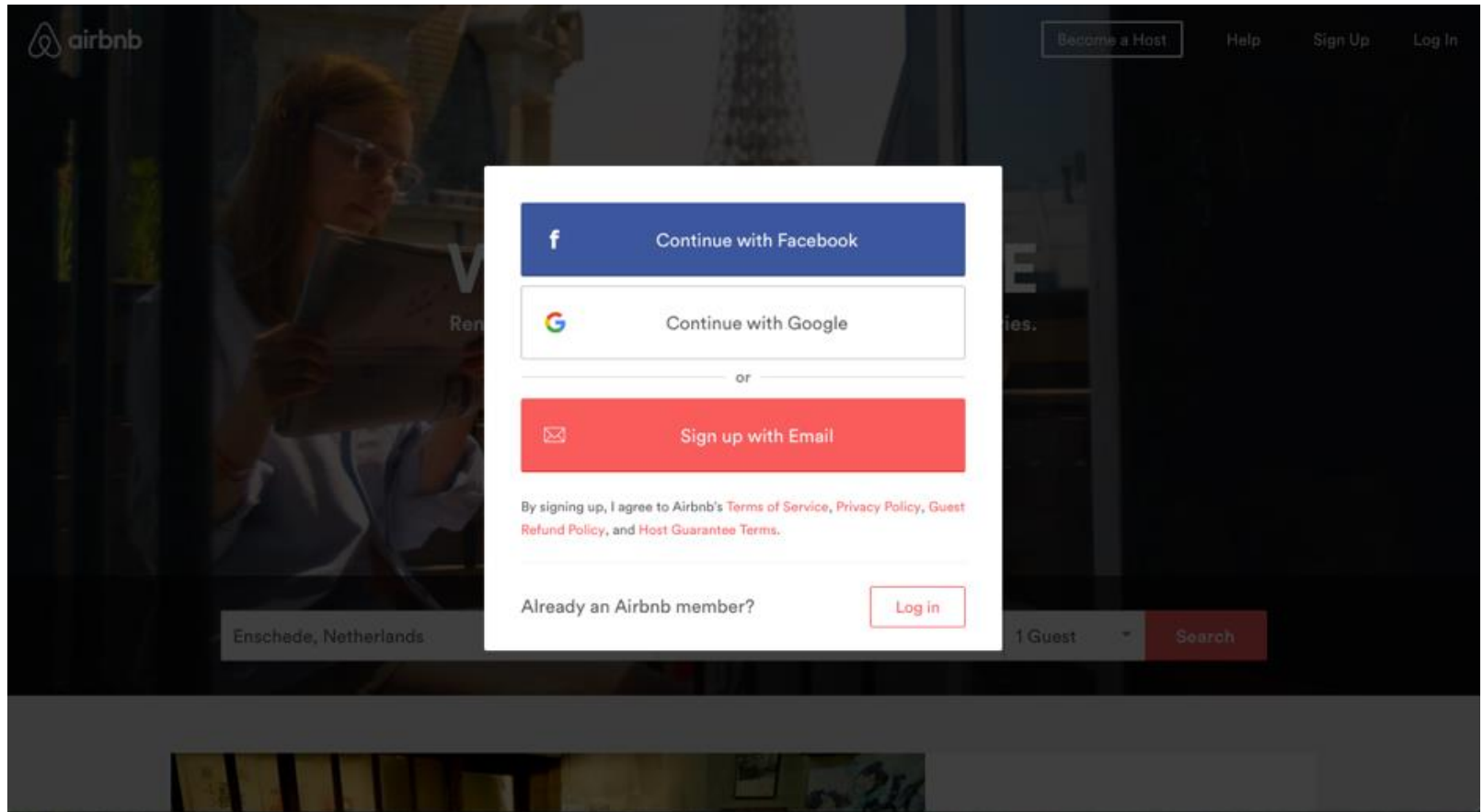


Figure and Ground

- Applied regularly for forms to login or register to a website.

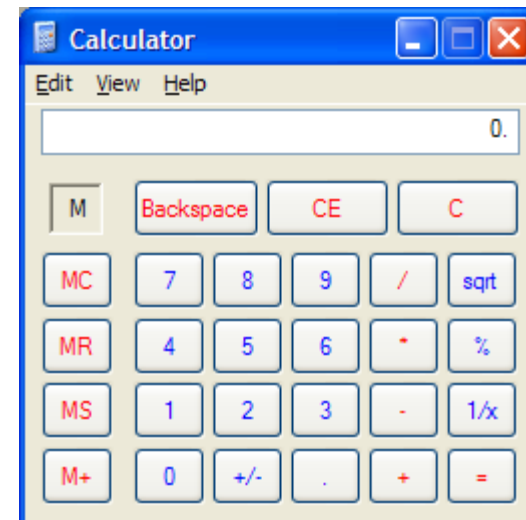


Creating Organization and Structure

- Grouping
- Hierarchy
- Whitespace
- Balance

Grouping

- Group elements into higher order units
 - e.g. Newspapers have paragraphs, columns, sections, pages
- Use the Gestalt principles to create groups
- Reserve powerful techniques such as colour and connectedness for explicitly telling the user something



Hierarchy

- One kind of hierarchy is visual: use visual variables to make the more **important elements pop out**
- The structure of information on a page that communicates order, emphasis, and relationship.
- Create the hierarchy to support the intended reading sequence

Guiding the eye on the page so that it attends to different design elements in the order of their importance

Visual Hierarchy

- Guiding the eye on the page so that it attends to different design elements in the order of their importance.

How to Train Your Brain to Remember Almost Anything

Four techniques for storing knowledge you might otherwise forget

 Thomas Oppong October 16 · 5 min read ★

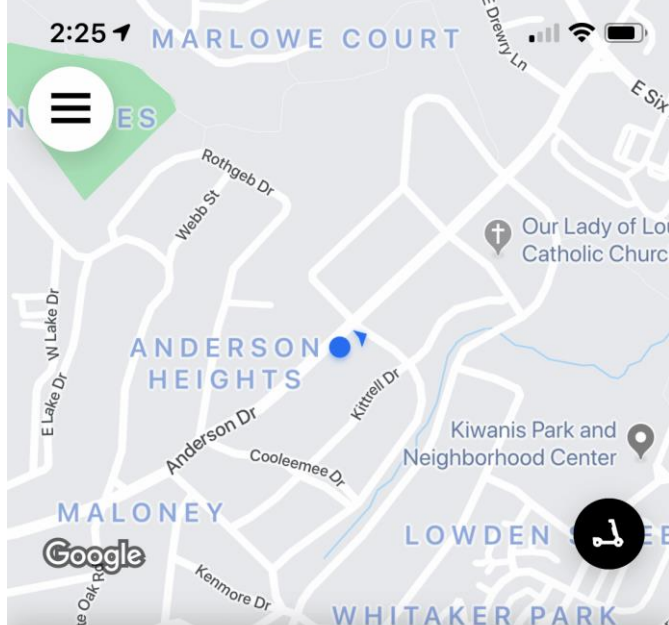


Photo: tunart/Getty

Success is largely based on what you know — everything you know informs the choices you make. And those choices are either getting you closer to what you want or increasing the distance between you and

Medium mobile app: There is a clear visual hierarchy of title, subtitle, and body text. Each component of the article is in a type size equal to its importance.





Good afternoon, Kelley

Where to?

Now



Home



Raleigh-Durham International Air...
2400 John Brantley Blvd, Morrisville



Blue

Your next reward



Ride



Order food

1:09



Home

Recent

Starred

Offline

OCTOBER 2019



00016794.pdf

Dropbox > housing

...



11 images

Dropbox > housing



GD_492

Dropbox

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GD400 Workshop Work

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MARCH 2019



House Reno Prep + Help Guides

Dropbox Paper

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Current House

Dropbox Paper

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FEBRUARY 2019



Kitchen Inspiration + Plans

Dropbox Paper

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OCTOBER 2018



Home



Files



+



Photos



Account

Visual Design

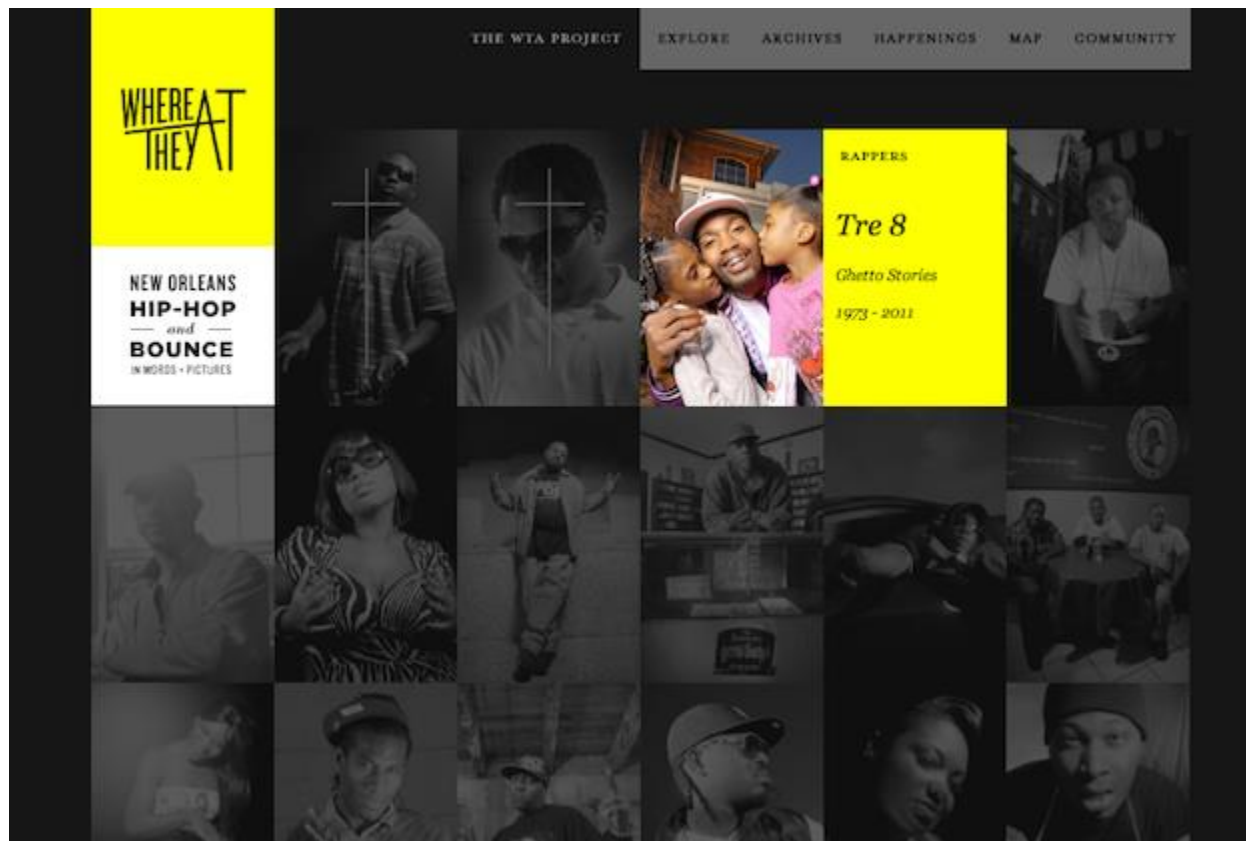
Visual Hierarchy

- Space and texture:
 - give content ample room to breathe
- Typeface weight and pairing
 - Typeface selection is critical to establishing visual hierarchy.



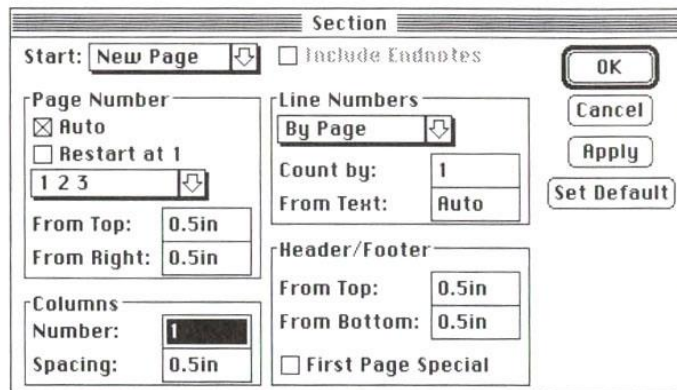
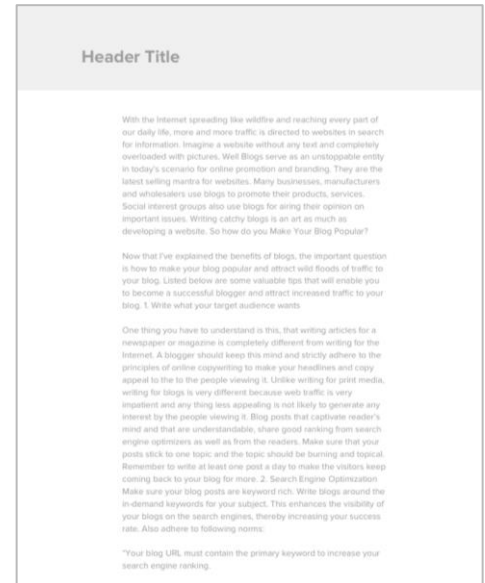
Visual Hierarchy

- Colors
 - bright colors stand out from muted colors or grayscale, while lighter tints appear more “distant” and thus fall lower on the hierarchy than richer, darker ones

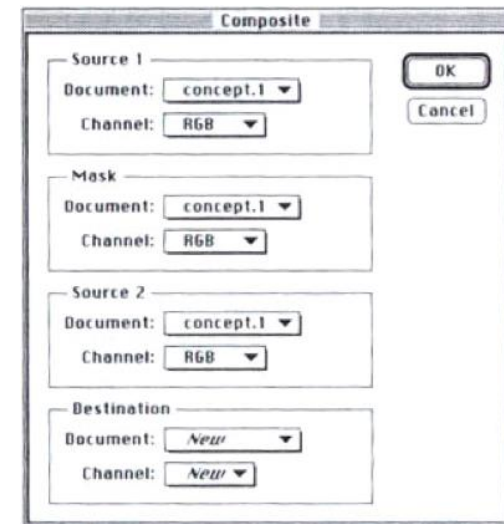


Whitespace

- Use white space for grouping, instead of lines
- Use margins to draw eye around design
- Integrate figure and ground
 - Object should be scaled proportionally to its background
- Don't crowd controls together
 - Crowding creates spatial tension and inhibits scanning



(a)



Hierarchy

- Start with structural hierarchy
 - how should the information be organized/grouped?
 - what's a title? heading? content? caption?
 - what's the same? what's different? what's related?
- Communicate levels/differences in hierarchy with one or more visual cues
 - whitespace, groupings, indents and alignments, line breaks
 - contrasting fonts, weights, colours or SMALL/LARGE CAPS
 - graphics, lines, rules, bars, etc.

Balance

- Try to create a stable composition by balancing elements, choose an axis (usually vertical)
- Distribute elements equally around the axis
 - Equalize both mass and extent



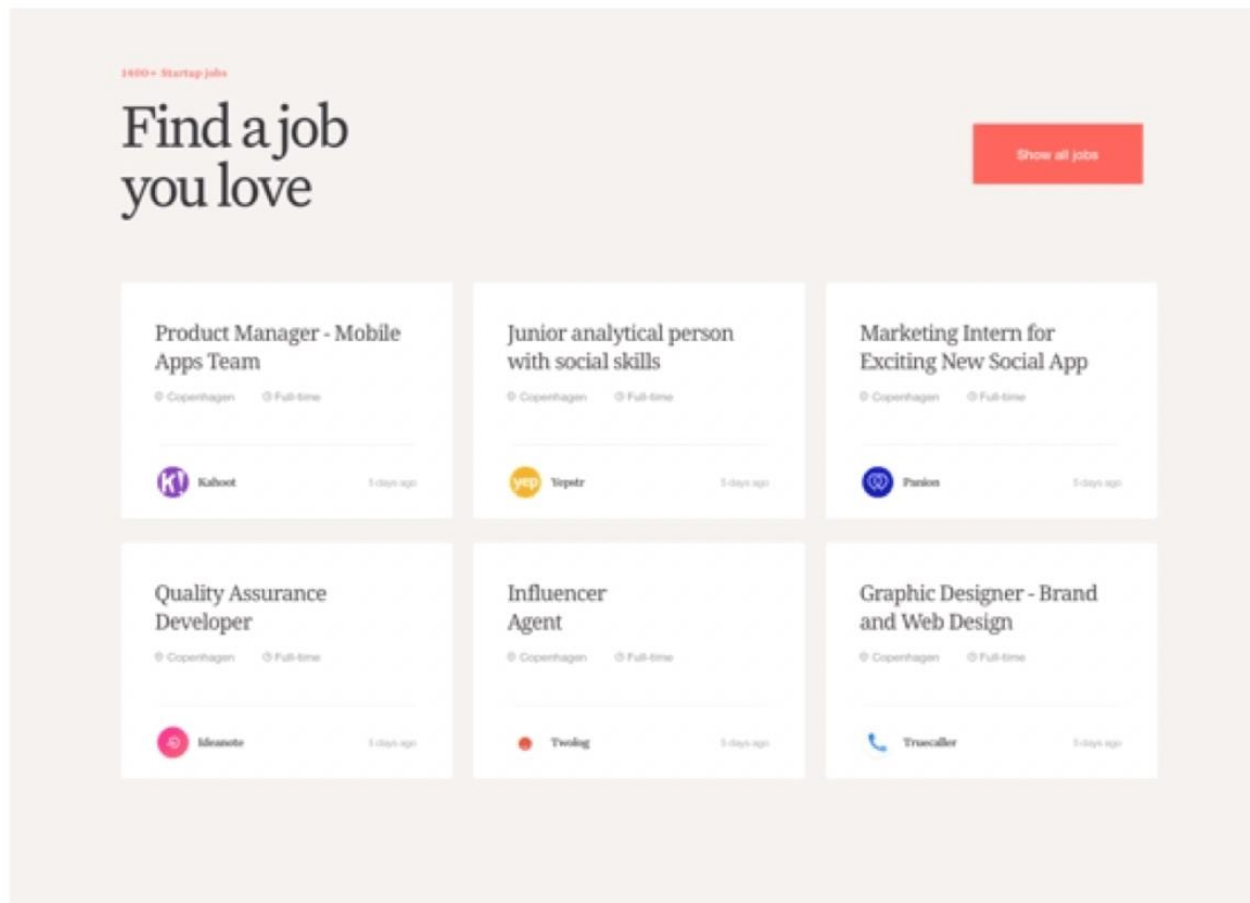
Imbalance



Balance

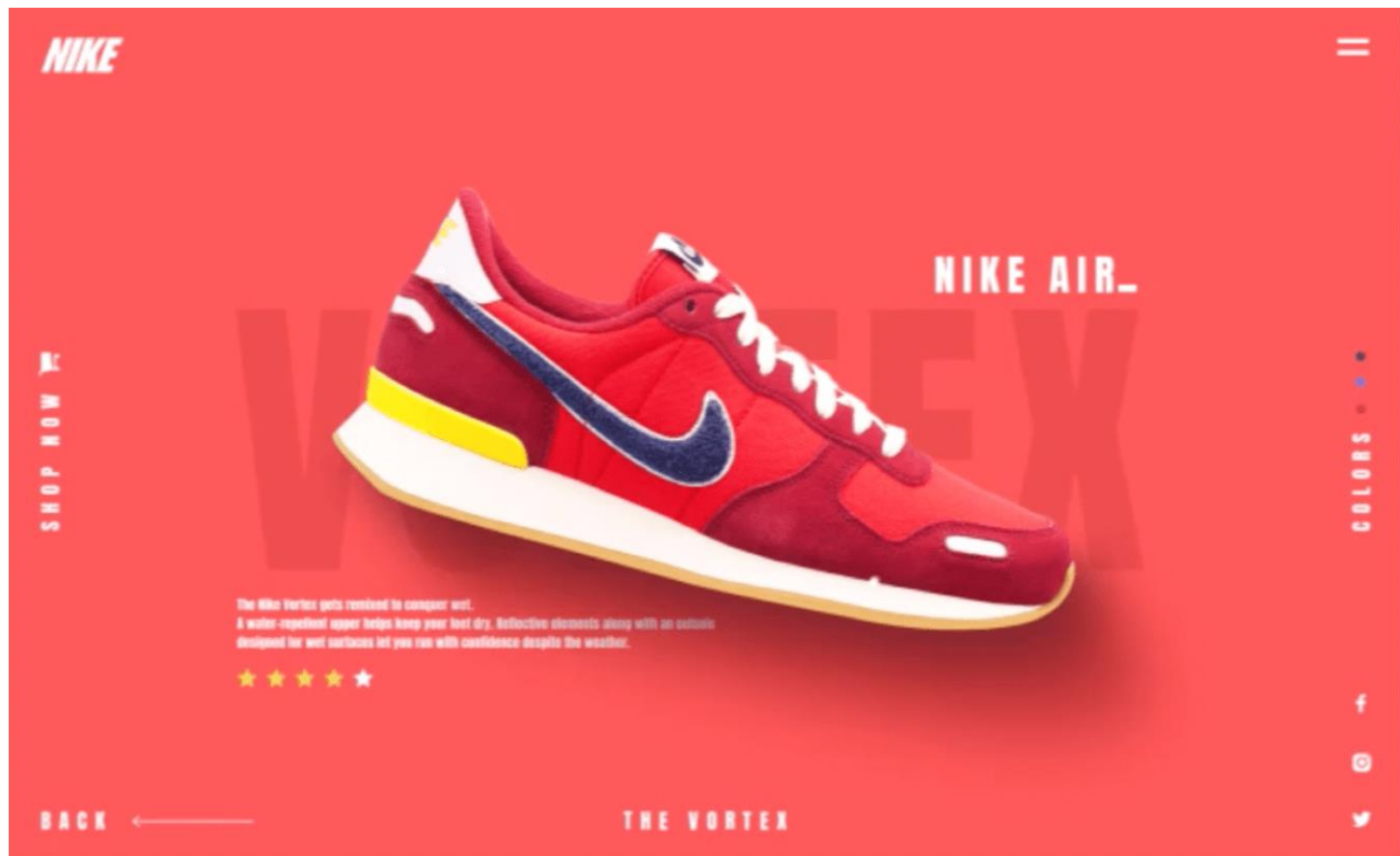
Balance

Symmetrical: elements are symmetrically distributed relative to the central imaginary axis



Balance

Asymmetrical: elements are asymmetrically distributed relative to the central axis



Typography

- The practice of arranging written subject matter
- Difference between “Typeface” and “Font”
- Typeface refers to a family of fonts sharing the same name, like “Arial” or “Georgia.”
- A font is a choice of typeface and size and style, like roman, italic, oblique, boldface, etc



The quick brown fox

The quick brown fox

The quick brown fox

The quick brown fox

Times New Roman

The quick brown fox

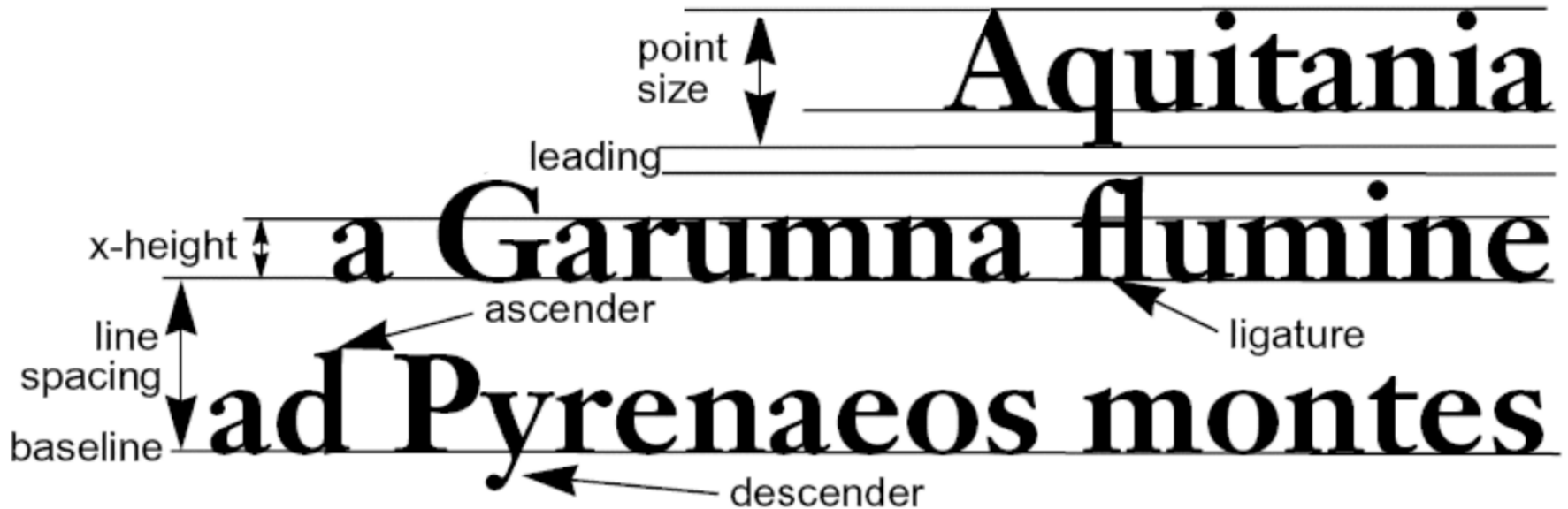
The quick brown fox

The quick brown fox

The quick brown fox

Open Sans

Terminology



More information: <https://www.canva.com/learn/typography-terms/>

Point size and Line Spacing

- A font is often measured in pt (points).
- Points dictate the height of the lettering.
 - There are approximately 72 (72.272) points in one inch or 2.54 cm.
- Line spacing (leading): measured from baseline to baseline
- Standard spacing = 120% of point size – e.g., 10pt font would have 12pt spacing

Designers play with line spacing in order to create distinctive layouts. Reducing the standard distance creates a denser typographic color – while risking collisions between ascenders and descenders. . . Expanding the line spacing creates a lighter, more open text block.

what is the effect of the different spacing?

Designers play with line spacing in order to create distinctive layouts. Reducing the standard distance creates a denser typographic color – while risking collisions between ascenders and descenders. . . . Expanding the line spacing creates a lighter, more open text block.

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Legibility vs. Readability

- Legibility: how well you can make out individual letters in a font
 - largely influenced by the design of the font itself
 - e.g., x-height, ascenders/descenders, ornament, etc.
- Readability: ease of understanding whole groups of text
 - influenced by the arrangement of whole groups of text
 - e.g., line spacing, measure

Type Styles

- Style (Sans-Serif, Serif, Display, etc)
 - Serifs are the little handles on some fonts, a decorative stroke that finishes off the end of a letters stem
 - Sans-serifs: literally without the serifs



VOGUE



COACH



hulu

LinkedIn

J.P.Morgan



SONY



citi

TARGET

T-Mobile



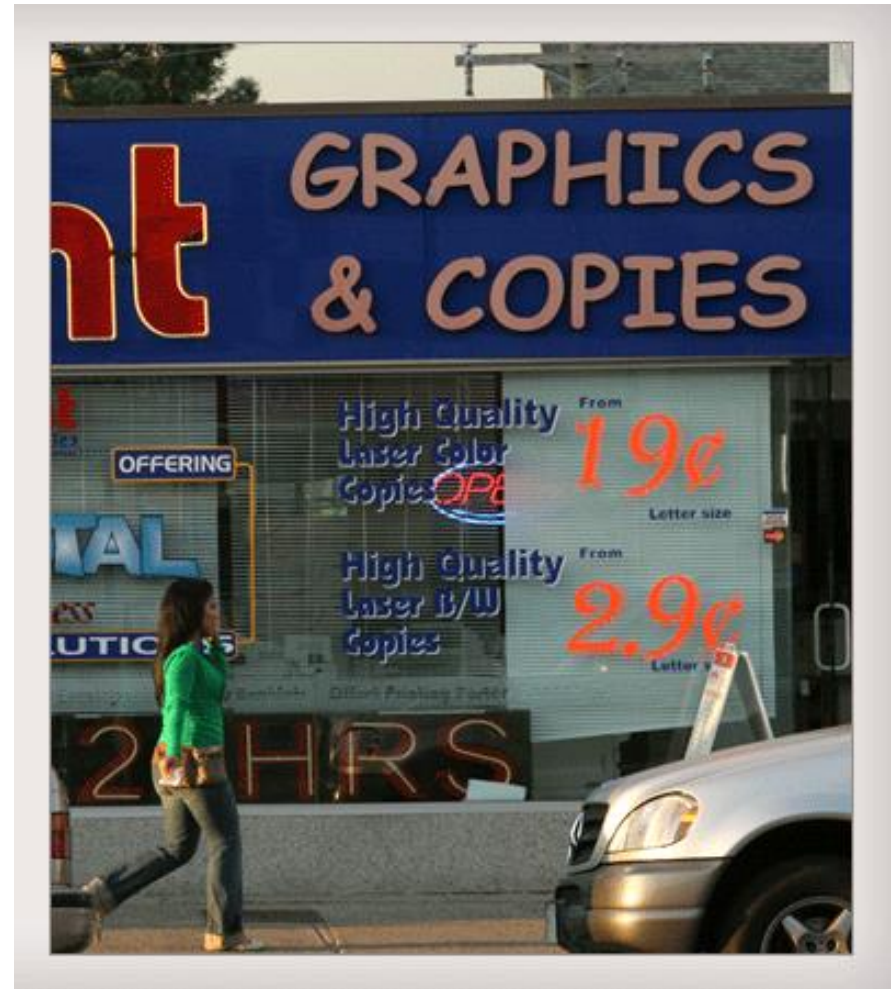
★ macy's

Guidelines for Using Sans-serif and Serif Fonts

- When to use which is a heavily debated question!
 - very few recent scientific studies
 - many old studies now criticized as flawed
- At small sizes
 - san-serif works better for body text while serif is appropriate for paragraph headers
- At standard and larger sizes
 - serif can be for body text (as the font is more legible and causes less eye strain or fatigue) when used to display longer, expansive passages

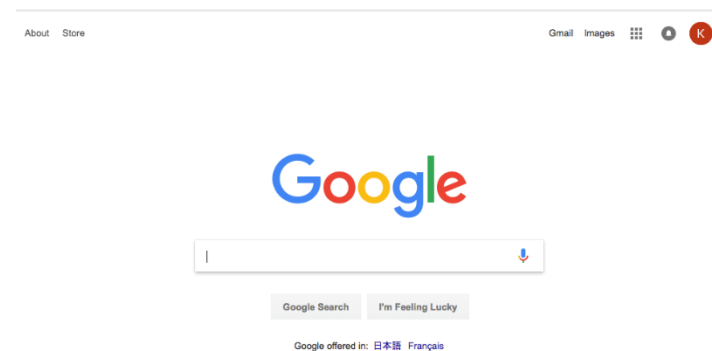
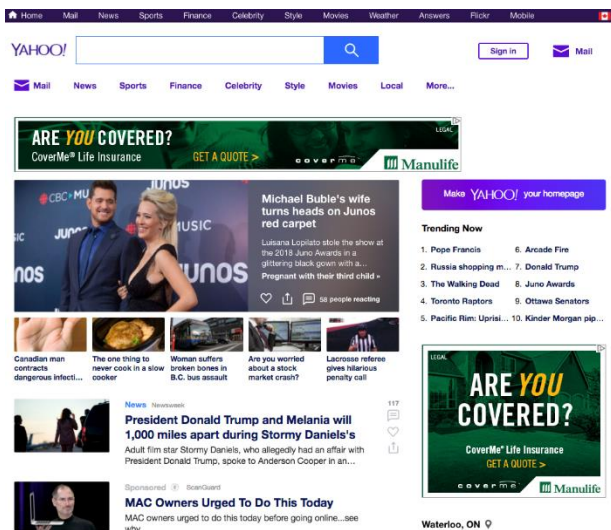
Some Guidelines

- Maximum 3 fonts
- Avoid using display typefaces like **comic sans**
- Consider using **bright colors** for important items and **muted colors** for less important ones.
- Incorporate **various scales** for your different design elements. A general rule of thumb is to include small, medium, and large components in the design.

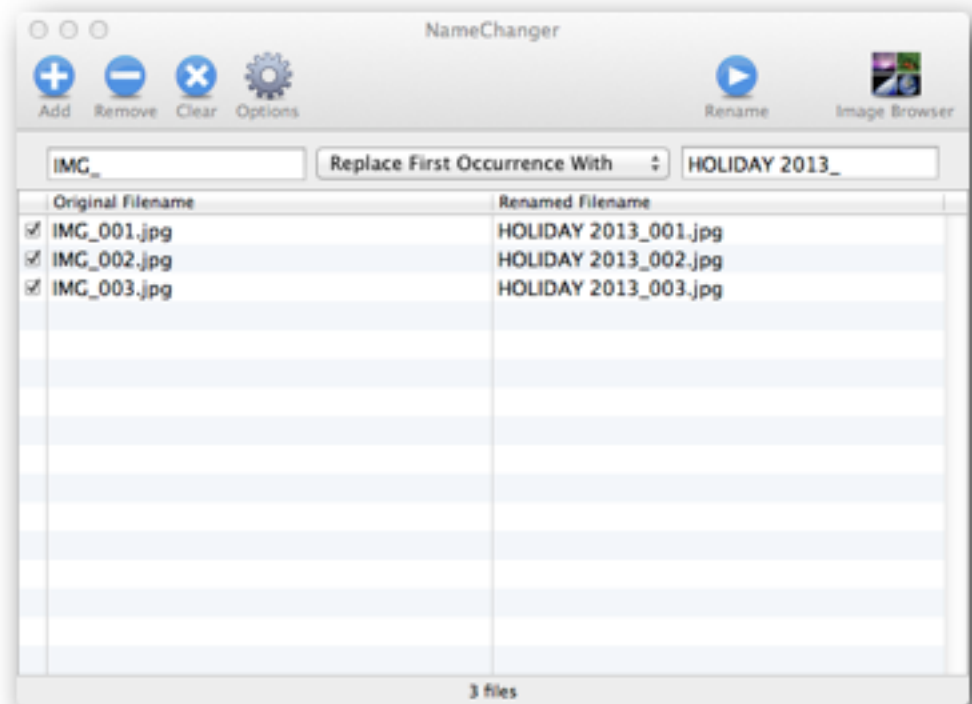
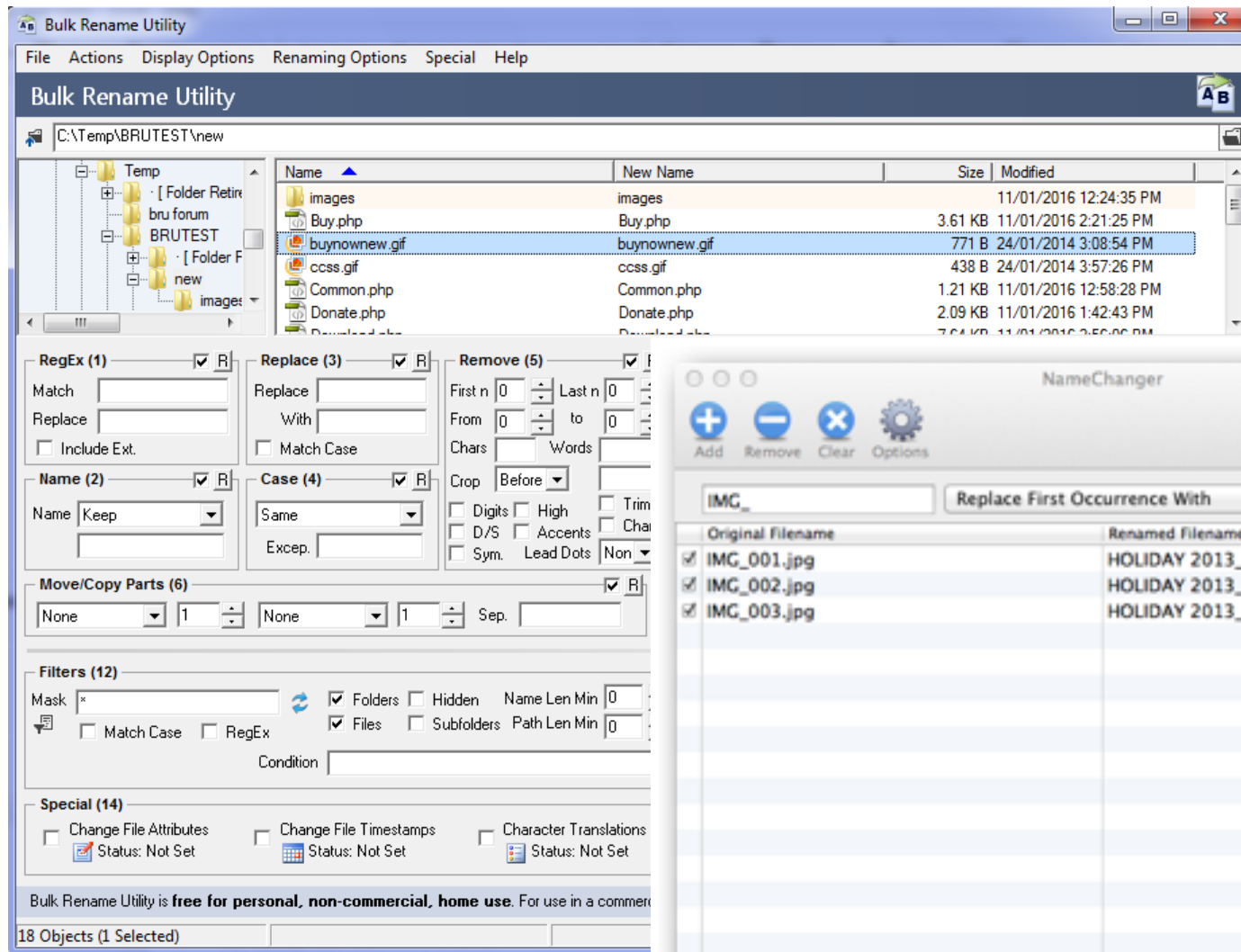


Simplicity

- Present the minimum amount of information to achieve maximum effect
- Functions are **quickly recognized and understood**
 - Less information means less time to process
 - Can more quickly produce correct mental models
- Simplicity also aids recall
 - Less to remember, easier to recall



What are the Essential Features?



How to achieve simplicity?

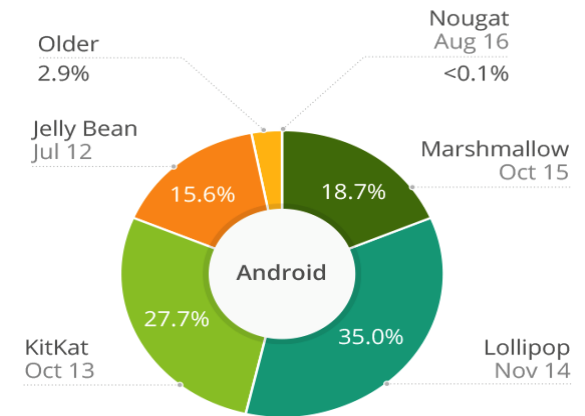
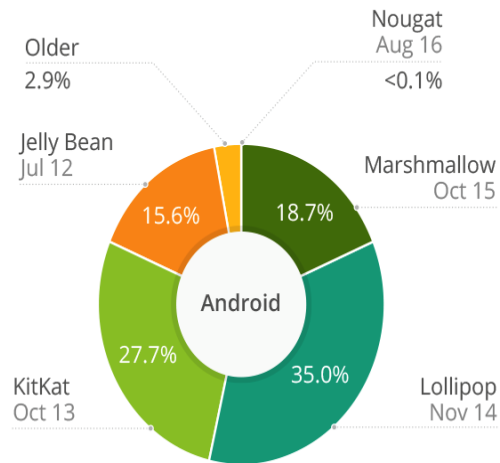
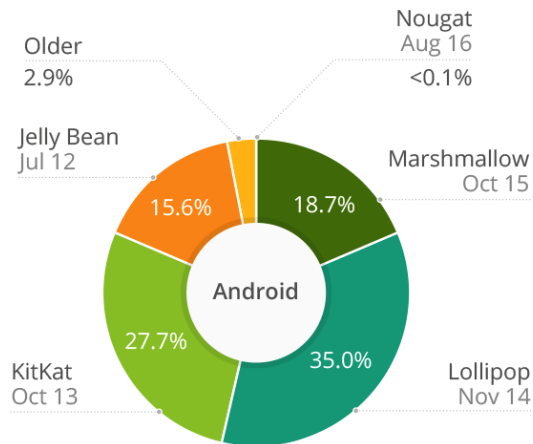
- Reduce, reduce, reduce
- Reduce some more
- Reduce until it hurts

Course Search: This is where you can search for courses based on a number of criteria including day and time offered, subject area and keywords in the course description. The results will display at the bottom of the page.

Search Criteria	Results per Page
1 Subject Area: eg. ENGL or EN* (eg. ENGL or EN*) 2 Course Number: eg. 110 or 1* (eg. 110 or 1*) 3 Keyword(s): 4 Activity Type: --Any-- 5 Credits: 6 Term: --Any-- 7 Status: --Any-- 8 Available Times: - (24 Hr format) 9 Available Days: ☐ Su ☐ M ☐ T ☐ W ☐ Th ☐ F ☐ Sa 10 Check here if Available days must match precisely ☐ 11 Return courses eligible for Credit/D/Fail only ☐ 12 Return Distance Education courses only ☐ Search for Sections	☒ 20 Items ☐ 50 Items ☐ 100 Items

Aspect Ratio

- Never change the **aspect ratio** of an image, illustration, or type



don't do this

Impact

- Good visual design can reduce human processing time
- Tullis redesigned lodging information screens (1984)
 - 5.5 vs. 3.2 sec avg search times

Pennsylvania

Bedford Motel/Hotel: Crinoline Courts
(814) 623-9511 S: \$18 D: \$20
Bedford Motel/Hotel: Holiday Inn
(814) 623-9006 S: \$29 D: \$36
Bedford Motel/Hotel: Midway
(814) 623-8107 S: \$21 D: \$26
Bedford Motel/Hotel: Penn Manor
(814) 623-8177 S: \$18 D: \$25
Bedford Motel/Hotel: Quality Inn
(814) 623-5188 S: \$23 D: \$28
Bedford Motel/Hotel: Terrace
(814) 623-5111 S: \$22 D: \$24
Bradley Motel/Hotel: De Soto
(814) 362-3567 S: \$20 D: \$24
Bradley Motel/Hotel: Holiday House
(814) 362-4511 S: \$22 D: \$25
Bradley Motel/Hotel: Holiday Inn
(814) 362-4501 S: \$32 D: \$40
Breezewood Motel/Hotel: Best Western Plaza
(814) 735-4352 S: \$20 D: \$27
Breezewood Motel/Hotel: Hotel 70
(814) 735-4385 S: \$16 D: \$18

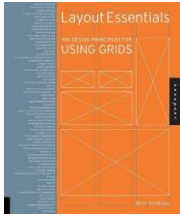
South Carolina

City	Motel/Hotel	Area Code	Phone	Rates	
				Single	Double
Charleston	Best Western	803	747-0961	\$26	\$30
Charleston	Days Inn	803	881-1800	\$18	\$24
Charleston	Holiday Inn N	803	744-1621	\$36	\$46
Charleston	Holiday Inn SW	803	556-7100	\$33	\$47
Charleston	Howard Johnsons	803	524-4140	\$31	\$36
Charleston	Ramada Inn	803	774-8281	\$33	\$40
Charleston	Sheraton Inn	803	744-2401	\$34	\$42
Columbia	Best Western	803	796-9400	\$29	\$34
Columbia	Carolina Inn	803	799-8200	\$42	\$48
Columbia	Days Inn	803	736-0000	\$23	\$27
Columbia	Holiday Inn NW	803	794-9440	\$32	\$39
Columbia	Howard Johnsons	803	772-7200	\$25	\$27
Columbia	Quality Inn	803	772-0270	\$34	\$41
Columbia	Ramada Inn	803	796-2700	\$36	\$44
Columbia	Vagabond Inn	803	796-6240	\$27	\$30

Summary

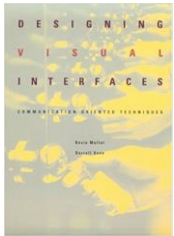
- Strive for simplicity
- Gestalt principles hint at how brain reacts to visual stimulus
- Use principles to structure a visual design by:
 - Grouping visual information into higher units
 - Creating a hierarchy to guide the user's viewing
 - Establishing relationships between elements
 - Creating a sense of balance

Recommended Books



Beth Tondreau, *Layout Essentials: 100 Design Principles for Using Grids* (2011)

- more print layout than interfaces, but excellent description of core layout and typography concepts



Kevin Mullet and Darrell Sano, *Designing Visual Interfaces: Communication Oriented Techniques* (1994)

- dated examples but concepts hold true; out of print, used copies can be cheap



Robert Bringhurst, *The Elements of Typographic Style* (2004)

- classic text on typography



William Lidwell, Kritina Holden, and Jill Butler, *Universal Principles of Design* (2010)

- high level overview of visual design concepts

Question?